

Project Proposal Template

Project: MindSight

Objective: Analyze a synthetic mental health dataset (50,000 records) to identify key risk factors. Deliver an interactive Power BI dashboard to support data-driven mental health decision-making.

Description: Dataset of 50,000 records with 15 features covering demographics, lifestyle, work conditions, and a binary target (Mental_Health_Condition). The pipeline covers data cleaning, data modeling, EDA, and an interactive dashboard.

Tools & Design: Power BI (dashboard & DAX), Power Query (data cleaning), SQL (data engineering). Designed around data storytelling and interactive visualization.

Technologies to Use: Data Cleaning & Star Schema Modeling, Feature Importance Analysis, Interactive Dashboard Development.

Team Members & Responsibilities:

- 1- Malak Ahmed Helmy Ibrahim
 - 2- Maryam Ahmed Elsayed
 - 3- Mayar Abdelrassoul Mahmoud
 - 4- Malak Ibrahim Abdo
 - 5- Rawan mousa alwani
 - 6- Hana Elgawish
-

Week 1: [Business Understanding (Hana Mohamed Elgawish)]

- Define project scope and business objectives
- Write the problem statement and business questions
- Create the data dictionary and define KPIs
- Identify key stakeholders and success criteria
- Research mental health risk factors and industry context
- Set the project timeline and milestones

Deliverables:

- Problem Statement Document
- Business Questions & KPIs Document
- Data Dictionary
- Project Timeline & Milestones Plan

Week 2: [Data Engineering (Malak Ahmed Helmy Ibrahim & Maryam Ahmed Elsayed)]

- Design the mini data pipeline (Raw → Quality → Feature Engineering)
- Perform data quality checks and validation
- Handle missing values and outliers
- Standardize data types and encode categorical fields
- Create engineered features (scores and indexes)
- Document the pipeline and data quality rules

Deliverables:

- Data Pipeline & Data Quality Report
 - Data Cleaning & Validation Log
 - Feature Engineering & Prepared Dataset
-

Week 3: [Data Cleaning & Data Modeling – Mayar Abdelrassoul Mahmoud]

- Import dataset into Power BI
- Clean and transform data using Power Query
- Encode categorical variables
- Build the Star Schema data model
- Create relationships between tables
- Develop DAX measures and KPIs

Deliverables:

- Clean Power BI dataset
- Completed data model with DAX measures

Week 4: [Dashboard Development & Storytelling – Malak Ibrahim Abdo & Rawan mousa alwani]

- Design dashboard layout and report pages
- Build interactive visuals and filters
- Add key insights and storytelling
- Publish the Power BI report
- Prepare final presentation

Deliverables:

- Final Power BI Dashboard
- Final Project Presentation